06. Symmetry

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. 4

2. 5

3. C — 3

4. 2

5. 1

6. 1

7. B — X

8. 6

9. 1

10. A, C and D (a scalene triangle and a non-rectangle/rhombus parallelogram have no lines of symmetry)

11. (3, -8)

12. (5, 2)

13. 2

14. 8

15. 1

Standard — Answers

16. 4

17. 3

18. D — 6

19. 2

20. Order 7, 7 lines of symmetry

21. C — parallelogram (not a rectangle or rhombus)

22. (1, 6)

23. (-4, -7)

24. A, B, D and E (H, N, S and Z all look the same after a half turn; F does not)

25. (-2, -3)

26. 30 cm² (15 + 15)

27. 9

28. A — X

29. (0, -5)

30. (6, 2)

31. 8

32. A, B and C (D is false — a non-square rectangle has only 2 lines of symmetry, not 4)

33. (7, -2)

34. 1

35. 10 lines of symmetry, order 10 (for any regular polygon, both equal the number of sides)

Hard — Answers

36. 48 cm² (the two lines of symmetry split the shape into 4 identical quarters: 4 × 12 = 48)

37. (-3, -4) (reflecting (3,4) in the x-axis gives Q = (3,-4); reflecting Q in the y-axis gives R = (-3,-4))

38. 12 sides, 12 lines of symmetry (for a regular polygon, the order of rotational symmetry always equals the number of sides, which always equals the number of lines of symmetry)

39. A, C and D (B is false — this kind of parallelogram has 0 lines of symmetry, not 2)

40. (-2, -5) (reflecting (5,-2) in y = x swaps the coordinates to give A = (-2,5); reflecting A in the x-axis gives B = (-2,-5))

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